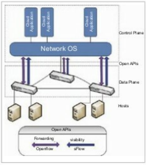

Figure 1: SDN architecture

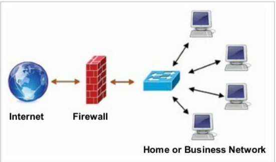

Figure 3: Traditional firewall

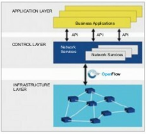

Figure 2: SDN layer architecture

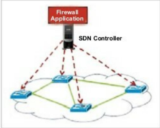

Figure 4: SDN based firewall

- Centrally managed
- Programmatically configured
- Experimenting and research is not expensive
- Fast upgrades

SDN architecture

SDN architecture is basically focused on allowing network administrators to manage and control the whole network through a software program based controller. This goal is achieved through the separation of the data plane and control plane, which simplifies the networking services.

Traditional vs SDN based firewalls

Here are a few differences between traditional and SDN based firewalls:

- Internal traffic is not seen and cannot be filtered by a traditional firewall.
- An SDN based firewall works both as a packet filter and a policy checker.
- The first packet goes through the controller and is filtered by the SDN firewall.
- The subsequent packets of the flow directly match the flow policy defined in the controller.
- The firewall policy is centrally defined and enforced at the controller.

Implementing an SDN based firewall

A firewall is used as a barrier to protect networked computers by blocking malicious network traffic generated by viruses and worms.

In this article, I have implemented an SDN based firewall by writing code for an SDN controller in Python. More specifically, I have written code to add firewall rules to a POX controller.

The implementation of an SDN based firewall requires the installation of Mininet and the POX controller. Both are open source tools and are freely available. For the installation and startup, please refer to my article at the following URL: *http://opensourceforu.com/2015/10/mininet-an-emulator-for-prototyping-large-network-topologies-on-a-single-machine/*. This article helps you to understand and work with Mininet. It emphasises the creation of a simple network topology using POX in Python.

In this article, I have implemented a layer 2 firewall that runs alongside the physical address based module on POX runtimes. The firewall application is provided with a list of MAC address pairs, i.e., an access control list (ACL). When a connection gets established between the controller and the switch, the application installs static flow rule entries in the OpenFlow table to disable all communication between each MAC pair.

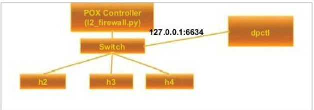

Figure 5: Firewall scenario in SDN

Examples of a few firewall rules

At the specified switch, block all traffic coming from host 10.1.1.1:

```
00-00-02-00-00-00-00-00 BLOCK srcip=10.1.1.1
```

At the specified switch, block all traffic coming from host 10.1.2.2, if the packet's TOS is marked with 32 and it's destined for 10.1.3.1:

```
00-00-03-00-00-00-00-00 BLOCK srcip=10.1.2.2 dstip=10.1.3.1
tos=32
```

At the specified switch, redirect traffic destined for 10.1.2.1 and, instead, send it to 10.1.2.2:

```
00-00-01-00-00-00-00-00 REDIRECT dstip=10.1.2.1 TO 10.1.2.2
```

At the specified switch, mark the TOS field of all packets sent by 10.1.1.1 with value 40:

```
00-00-04-00-00-00-00-00 MARK srcip=10.1.1.1 TOS 40
```